SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**
Question Count: 5
Duration :60 Minutes

Code of Conduct

I M.Kyathi Sai Priya(192572101)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Kyathi Sai Priya

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. **Agriculture Smart Farming Problem**
A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated

from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

1. Retail Data Optimization Problem

A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

Answer:

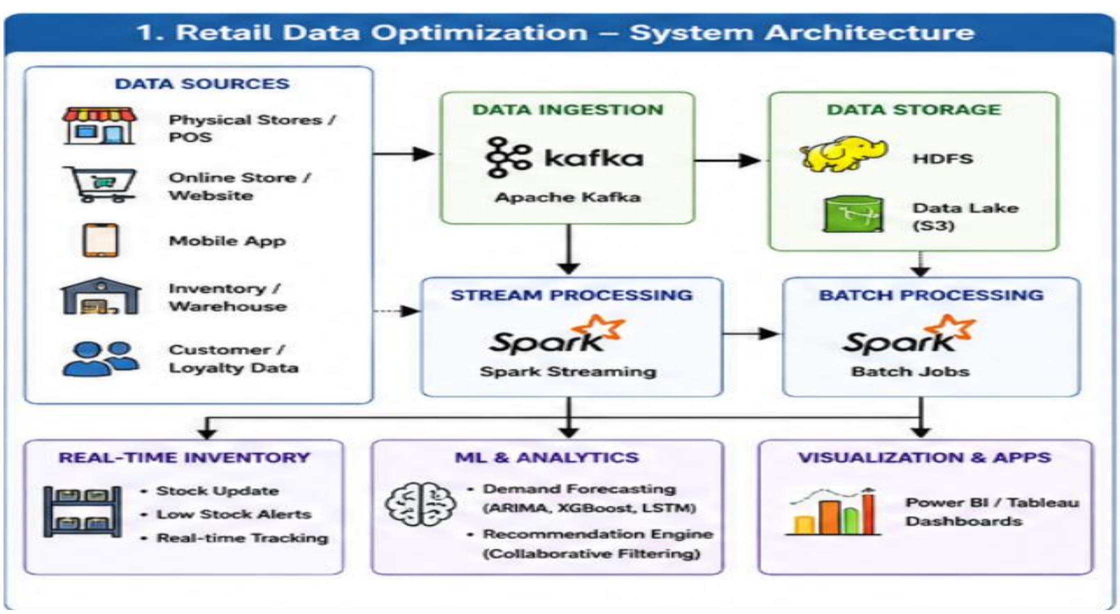
A big data architecture can integrate data from physical stores, online platforms, inventory systems, websites, and customer applications.

Architecture:

Data Sources → Kafka → Data Lake/HDFS → Spark Processing → Machine Learning → Dashboard/Applications

- Data Sources: POS transactions, online orders, inventory, customer browsing, mobile apps, and loyalty programs.
- Data Ingestion: Apache Kafka can collect real-time sales and customer activity.
- Storage: Hadoop HDFS or cloud data lakes can store large volumes of structured and unstructured data.
- Processing: Apache Spark can perform batch and real-time processing.
- Inventory Tracking: Streaming data can update stock levels immediately and generate low-stock alerts.
- Demand Forecasting: ARIMA, Random Forest, XGBoost, or LSTM models can predict future product demand.
- Recommendations: Collaborative filtering and content-based recommendation algorithms can provide personalized products.
- Visualization: Power BI, Tableau, or Grafana can display sales, inventory, and forecasting dashboards.

Benefits: Real-time inventory visibility, reduced stock-outs, better demand prediction, personalized marketing, and improved customer satisfaction.



2. Social Media Fraud Detection Problem

Question:

A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

Answer:

A scalable real-time architecture can continuously analyze social media activities and identify suspicious behavior.

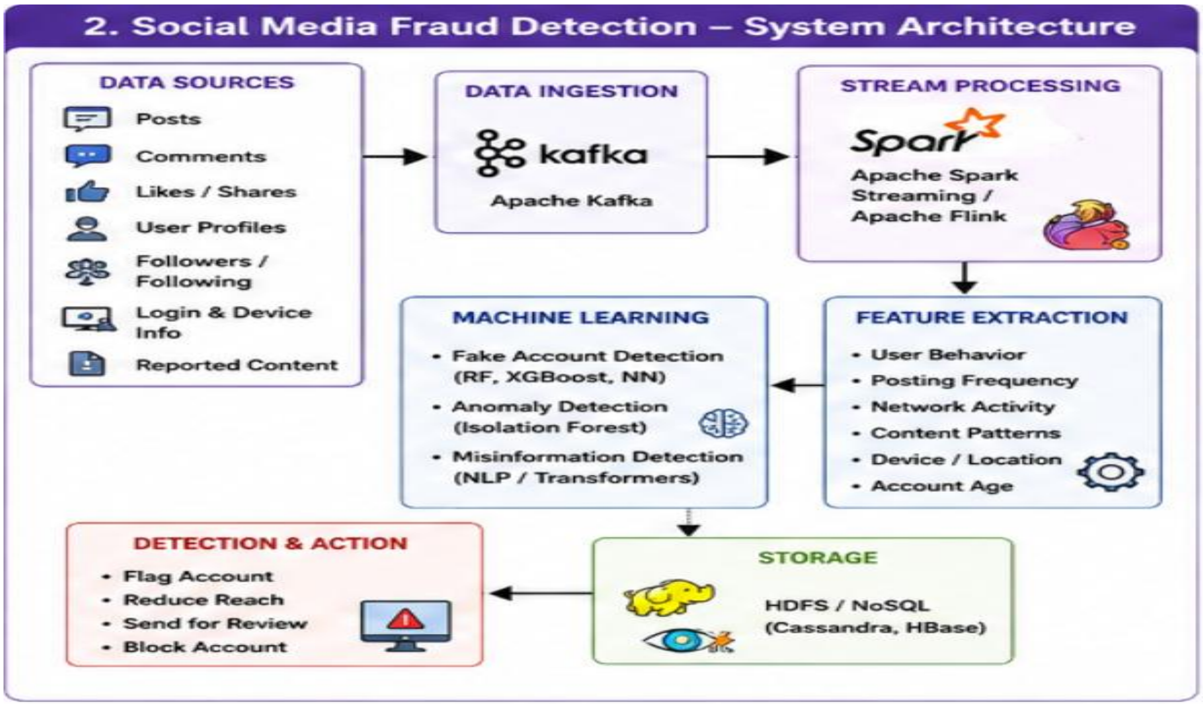
Architecture:

Social Media Events → Kafka → Spark Streaming/Flink → Feature Extraction → Machine Learning → Alerts/Actions

- Data Sources: Posts, comments, likes, shares, user profiles, followers, devices, and login activities.
- Data Ingestion: Apache Kafka collects millions of events continuously.
- Stream Processing: Apache Spark Streaming or Apache Flink processes events in real time.
- Features: Posting frequency, account age, follower ratio, repeated content, login patterns, and sharing behavior.
- Fake Account Detection: Random Forest, XGBoost, Logistic Regression, and neural networks can classify fake accounts.
- Anomaly Detection: Isolation Forest and clustering can identify unusual user behavior.
- Misinformation Detection: NLP and transformer-based models can analyze text and classify suspicious content.
- Graph Analytics: User relationships can be represented as graphs to identify coordinated fake-account networks.
- Storage: HDFS or NoSQL databases such as Cassandra can store historical data.

- Alerts: Suspicious accounts can be flagged, restricted, or sent for human review.

Benefits: Fast fraud detection, scalable processing, detection of coordinated attacks, and improved platform security.



3. Government Data Platform Problem

Question:

A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

Answer:

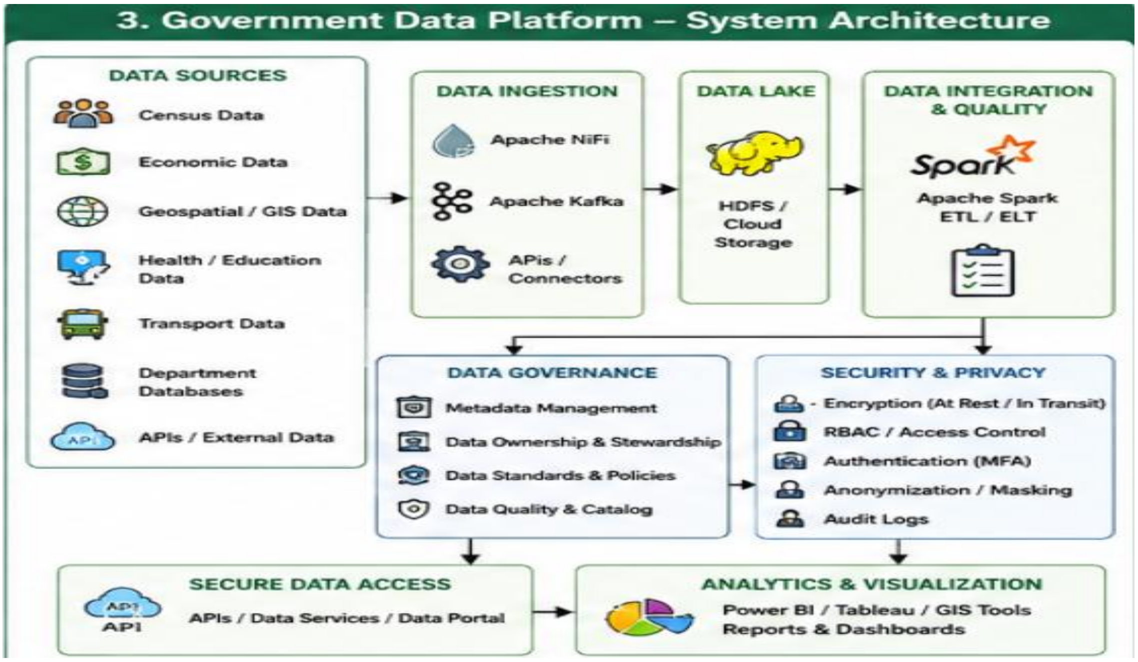
A national government data platform should provide centralized data integration while maintaining strong governance, security, privacy, and interoperability.

Architecture:

Government Departments → Data Ingestion → Data Lake → Data Integration → Governance → Secure Access → Analytics

- Data Sources: Census, economic, taxation, healthcare, education, transportation, and geospatial datasets.
- Storage: HDFS or cloud-based data lakes can store large datasets.
- Data Integration: Apache NiFi, Kafka, Spark, and APIs can transfer and integrate data.
- Interoperability: Common data formats such as JSON, XML, and CSV and standardized APIs should be used.
- Metadata: Every dataset should contain information about its source, owner, meaning, and quality.
- Data Governance: Define data ownership, stewardship, quality standards, access policies, and retention rules.
- Data Quality: Remove duplicates, correct errors, handle missing values, and validate formats.
- Security: Use encryption, role-based access control, multi-factor authentication, and audit logs.
- Privacy: Sensitive information should use anonymization, pseudonymization, and data masking.
- Analytics: Spark, Python, R, Power BI, and GIS tools can support government decision-making.

Benefits: Better data sharing, interoperability, improved data quality, stronger privacy, and efficient government services.



4. Banking Fraud Detection Problem

Question:

A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

Answer:

A real-time banking fraud detection system should combine streaming technologies, historical data, and machine learning.

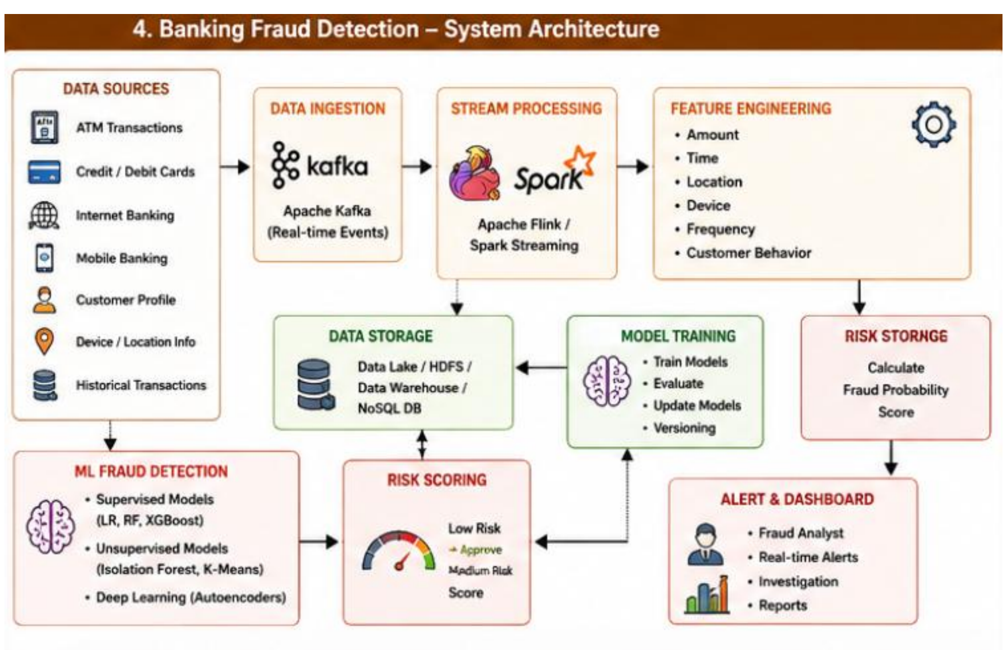
Architecture:

Transactions → Kafka → Flink/Spark Streaming → Feature Engineering → ML Model → Risk Score → Alert/Block → Data Lake

- Data Sources: ATM, credit/debit cards, mobile banking, online banking, customer profiles, and transaction history.
- Data Ingestion: Apache Kafka continuously receives transaction events.

- Stream Processing: Apache Flink or Spark Streaming analyzes transactions with low latency.
- Features: Transaction amount, location, time, device, transaction frequency, and customer spending pattern.
- Supervised Algorithms: Logistic Regression, Decision Tree, Random Forest, and XGBoost can classify transactions as genuine or fraudulent.
- Unsupervised Algorithms: Isolation Forest, K-Means, and Autoencoders can detect unknown fraud patterns.
- Risk Scoring: Each transaction receives a fraud probability score.
    - Low risk → Approve.
    - Medium risk → Additional authentication.
    - High risk → Block/hold and generate alert.
- Historical Storage: HDFS, cloud data lakes, NoSQL databases, or data warehouses can store transaction history.
- Dashboard: Power BI, Tableau, or Grafana can show fraud trends and alerts.
- Model Training: Models can be periodically retrained using new transaction data.

Benefits: Low-latency detection, high scalability, improved accuracy, reduced financial losses, and early identification of new fraud patterns.



5. Agriculture Smart Farming Problem

Question:

A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Answer:

A precision agriculture system can combine IoT sensors, drones, satellite data, weather information, and machine learning.

Architecture:

Sensors/Drones → IoT Gateway → Kafka → Data Lake → Spark → Machine Learning → Farmer Dashboard

- Data Sources: Soil moisture, soil temperature, pH, rainfall, weather, crop images, drones, satellites, and irrigation systems.
- Data Collection: IoT sensors continuously collect field information.
- Data Ingestion: Kafka can transmit real-time sensor data.
- Storage: HDFS or cloud storage can store sensor data, images, weather information, and historical records.
- Processing: Apache Spark can process large-scale agricultural datasets.
- Yield Prediction: Regression, Random Forest, XGBoost, and LSTM can predict crop yield.
- Irrigation Management: Soil-moisture data can determine when irrigation is required, reducing water wastage.
- Disease Detection: CNN-based deep learning models can analyze drone or crop images to detect diseases.

- Weather Analytics: Historical and real-time weather data can help farmers plan sowing, irrigation, and harvesting.
- Dashboard: Farmers can monitor soil moisture, crop health, weather, irrigation requirements, disease alerts, and predicted yield.

Benefits: Higher crop yield, reduced water and fertilizer usage, early disease detection, lower costs, and better data-driven farming decisions.